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## VIEWS OF THE ORION NEBULA

### 1 Orion's sword

The Orion Nebula is one of the most observed and photographed objects in the night sky. To the naked eye it is a just a fuzzy patch marking the sword of Orion. However, photographs transform it into a colorful maelstrom of star-birth. This wide-field view shows all of the massive star-formation region. It was taken by the VISTA infrared telescope at the European Paranal Observatory in Chile.

### 2 Heart of the nebula

The heart of the Orion Nebula stellar nursery contains thousands of young stars and developing protostars. The infant stars have blown away much of the dust and gas in which they formed, carving a huge cavity in the cloud, seen here in red. The brilliant starlight of the region at the top of the cavity comes from a tight open cluster of young stars, known as the Trapezium (see p.162).

### 3 Infrared composite

This infrared view combines data from two space telescopes, Spitzer and Herschel. It shows a region of the nebula about 10 light-years across, with the Trapezium to the left of the image. In infrared light, it is dust rather than the gas and stars that shines brightest. The red regions show cold dust, condensed into clumps around stars in the process of forming. Blue indicates warmer dust, heated by fully formed hot, young stars.

### 4 High-temperature gas

A high-temperature gas cloud, shown here in blue, is revealed when the nebula is imaged in X-ray light by the space telescope XMM-Newton. It appears to fill the nebula's huge cavity, which is visible in optical and infrared views. The cloud was produced in a violent collision when wind from a massive star was heated to millions of degrees as it slammed into surrounding gas. The bright yellow patch is the Trapezium cluster.